Factor-Based ML Long-Short Portfolio

1. Factors Constructed and Fundamental Features Used

Four fundamental factor categories were built from the supplied annual ratio data (2017-2022). Each category combines 3 sub-metrics into a cross-sectional percentile score (0-1, higher = more attractive), oriented so that a higher score is always more attractive regardless of whether the underlying raw metric is 'higher is better' or 'lower is better'. Category scores are simple averages of their sub-metric percentiles, and an overall composite fundamental score is the equal-weighted average of the four category scores.

Category	Sub-Metrics Used	Direction
Valuation	Price/Earnings, Price/Book, EV/EBITDA	Lower = better
Profitability	Rate of Return (ROE proxy), Gross Profit/Assets, Earnings Yield (Greenblatt)	Higher = better
Growth	YoY EPS Growth, YoY Revenue Growth, YoY EBITDA Growth	Higher = better
Leverage / Quality	Debt/Equity, Current Ratio, Asset Turnover	Mixed (D/E lower better; others higher better)

Table 1: Fundamental factor categories and their constituent sub-metrics.

Momentum (mandatory per the assignment brief) is constructed separately, directly from real daily price data, as 1/3/6/12-month trailing returns.

Missing-Data Handling

The supplied fundamentals file has real, non-trivial gaps: roughly 28-38% missing per metric in 2017 (limited historical backfill for some names), a low single-digit percentage in 2018-2021, and 75-85% missing in 2022 (a reporting-lag artifact — most FY2022 annual filings were not yet published when the data was compiled). A three-stage imputation cascade was applied to every metric, per stock, before any percentile ranking: (1) carry-forward the stock's own prior-year value; (2) sector median for values still missing (typically a stock's first year on file); (3) universe median as a last-resort fallback. All gaps were fully resolved after the sector-median stage; the universe-median fallback was never actually needed.

Metric	Missing (raw)	After carry-forward	After sector-median	Final missing
Price/Earnings	203	120	0	0
Price/Book	144	65	0	0
EV/EBITDA	133	57	0	0
Rate of Return (ROE proxy)	226	147	0	0
Gross Profit/Assets	108	32	0	0
Earnings Yield (Greenblatt)	124	48	0	0
YoY EPS Growth	135	44	0	0
YoY Revenue Growth	111	33	0	0
YoY EBITDA Growth	116	34	0	0
Debt/Equity	151	75	0	0
Current Ratio	108	32	0	0
Asset Turnover	109	33	0	0

Table 2: Missing-cell counts before and after each imputation stage (102 stocks x 6 years per metric).

Fundamental features are lagged one calendar year when merged into the monthly ML feature panel (FY-year Y ratios apply to months of calendar year Y+1, with 2017 as a boundary case using same-year data since no 2016 fundamentals exist) - this avoids look-ahead bias from real-world reporting lags.

2. Features and Model Used in ML-Based Ranking

Target variable: next-month return (%). Approach: regression (numeric return prediction; stocks are then ranked by predicted value, rather than classified into buckets).

Group	Features
Momentum (price-based)	mom_1m, mom_3m, mom_6m, mom_12m (trailing 1/3/6/12-month returns)
Fundamental category scores	valuation_score, profitability_score, growth_score, leverage_quality_score
Headline fundamental percentiles	pe_pctile, roe_proxy_pctile, eps_growth_pctile, de_pctile

Table 3: The 12 features used to predict next-month return.

Train/test split: Training = January 2017-December 2021 (~6,575 stock-months after dropping rows with missing targets/features, mostly each stock's first few months where trailing momentum cannot yet be computed); Test = January 2022 onward. Six regression models were

trained and compared: Linear, Ridge, Lasso, Random Forest, XGBoost, and LightGBM.

Model selection criterion: average monthly Rank Information Coefficient (IC) - the Spearman rank correlation between each model's predicted and realized next-month returns, averaged across test months. This is the standard metric for ranking-based strategies since it measures how well a model orders stocks by future return, rather than its point-estimate accuracy.

Model	RMSE	R^2	Avg. Monthly Rank IC
Linear Regression	0.1272	-0.1018	-0.0808
Ridge Regression	0.1272	-0.1017	-0.0809
Lasso Regression	0.1267	-0.0924	-0.0914
Random Forest	0.1295	-0.1417	-0.0483 (best / least negative)
XGBoost	0.1349	-0.2391	-0.0598
LightGBM	0.1343	-0.2284	-0.0674

Table 4: Out-of-sample RMSE, R^2, and average monthly Rank IC for each model, 2022 test period (real prices).

On real 2022 data, every model produced a negative average Rank IC and negative R^2 out of sample. Random Forest had the least-negative Rank IC (-0.0483) and was therefore selected as the 'best available' ranking model per the stated selection rule, and used for SHAP interpretation below - but this selection should not be read as evidence of genuine predictive skill; none of the six models beat a naive random ranking in this test window (see Results and Limitations, Section 4).

SHAP Feature Importance

SHAP (SHapley Additive exPlanations) values were computed on the selected Random Forest model to interpret which features drove its next-month return predictions.

Feature	Mean |SHAP value|
mom_3m (3-month momentum)	0.009753
mom_1m (1-month momentum)	0.006308
mom_12m (12-month momentum)	0.002643
valuation_score	0.001580

mom_6m (6-month momentum)	0.000858
de_pctile	0.000711
pe_pctile	0.000581
leverage_quality_score	0.000405
eps_growth_pctile	0.000366
profitability_score	0.000362
growth_score	0.000349
roe_proxy_pctile	0.000279

Table 5: Mean absolute SHAP value per feature. Short-to-medium-term momentum (3-month, 1-month, 12-month) dominates, followed by valuation score.

3. Method Used to Build the Final Ranking and Portfolio

At each bi-monthly rebalance date (Jan, Mar, May, Jul, Sep, Nov 2022), the selected model (Random Forest) produces a predicted next-month return for every stock in the 97-stock universe using that month's momentum and lagged fundamental features. Stocks are ranked by predicted return:

- Top 3 predicted stocks -> long leg
- Bottom 3 predicted stocks -> short leg
- Dynamic weights: weight_i = |predicted_return_i| / sum(|predicted_return|) within each leg — stronger-conviction signals receive proportionally larger allocations
- Equal-weight benchmark: same stock selection, but 1/3 weight per leg
- Positions are held for a full 2-month period, then closed; the next signal immediately opens the following 2-month position (non-overlapping, continuous coverage Feb-Dec 2022)

Both legs are fully dollar-neutral (100% long, 100% short, 0% net exposure, 200% gross).

4. Results

Monthly Portfolio Returns, 2022

Month	ML Dynamic-Weight	Equal-Weight
2022-02	-1.67%	-1.73%
2022-03	-6.58%	-6.48%
2022-04	-4.67%	-4.26%
2022-05	-10.40%	-10.19%
2022-06	+6.26%	+6.17%
2022-07	+3.53%	+3.43%
2022-08	+0.29%	+0.37%

2022-09	+4.65%	+4.68%
2022-10	-6.35%	-6.36%
2022-11	+7.03%	+7.27%
2022-12	-2.73%	-3.20%

Table 6: Realized monthly returns for both weighting schemes, real market data, 2022.

The two weighting schemes track each other very closely throughout the year - a direct consequence of the model's weak, near-random predicted-return spread: when a model's conviction (predicted return magnitude) barely differs across the selected stocks, |predicted_return|-weighting collapses toward equal weighting.

Performance and Fama-French 3-Factor Alpha

Monthly excess portfolio returns were regressed on the real FF3 factors downloaded from Kenneth French's data library: R_portfolio - R_f = alpha + beta_MKT_RF*(MKT-RF) + beta_SMB*SMB + beta_HML*HML + epsilon. All 11 available portfolio months (Feb-Dec 2022) matched real FF3 data with no gaps.

Metric	ML Dynamic-Weight	Equal-Weight
Total Return (11 months)	-11.7%	-11.3%
Annualized Return	-12.7%	-12.3%
Annualized Volatility	20.1%	20.0%
Sharpe Ratio	-0.58	-0.56
Max Drawdown	-20.2%	-19.6%
FF3 Monthly Alpha	0.17%	0.18%
FF3 Annualized Alpha	2.06%	2.21%
Beta: MKT-RF	-0.06	-0.06
Beta: SMB	-0.26	-0.30
Beta: HML	-0.91	-0.90
FF3 Regression R^2	0.51	0.49
Regression Observations	11 months (Feb-Dec 2022)	11 months (Feb-Dec 2022)

Table 7: Full performance and FF3-alpha summary, 2022 backtest, real data.

Interpretation and Limitations

- Both portfolios lost money in 2022 (-11.7% and -11.3% total return) with negative Sharpe ratios (-0.58 and -0.56) — the strategy did not generate positive absolute returns in this backtest window.
- Both portfolios carried a very strong negative Value (HML) beta (-0.91 and -0.90). 2022 was a year in which value stocks sharply outperformed growth/momentum stocks amid aggressive Fed rate hikes; the

portfolio's long side, tilted toward growth/momentum names, sat on the wrong side of this rotation. Most of the portfolio's loss is explained by this factor exposure rather than by stock-selection error.

- The FF3 regression alpha is small and technically positive for both portfolios (annualized +2.06% and +2.21%), so alpha > 0 in a literal, point-estimate sense — but this must be interpreted with real caution: the regression has only 11 monthly observations against 4 estimated parameters (7 degrees of freedom). With so few observations, standard errors on the alpha estimate are large, and a monthly alpha of ~0.17-0.18% is very unlikely to be statistically distinguishable from zero. The honest conclusion is that the strategy showed no reliable evidence of skill (positive or negative) after controlling for factor exposures, rather than a confirmed positive-alpha result.
- Every one of the six candidate ML models produced a negative average monthly Rank IC and negative out-of-sample R^2 in the 2022 test period — meaning none of them ranked stocks better than chance in this specific year. Random Forest was 'best' only in a relative, least-negative sense. This is a legitimate empirical finding, not a bug: 2022's sharp, macro-driven regime shift (rate-hike-triggered growth-to-value rotation) is exactly the kind of environment in which historically-trained momentum/fundamental factors can break down, and a single 11-month out-of-sample test window is a small, noisy sample to evaluate any model on.
- SHAP analysis shows short-to-medium-term momentum (3-month, 1-month, 12-month) as the dominant input to the selected model's predictions, well ahead of any fundamental feature — despite this, momentum-driven predictions did not translate into positive Rank IC in this test window.
- Recommended next steps if further work is done on this strategy: test across a longer, multi-year out-of-sample window (not just 2022) to see whether poor performance is specific to this year's regime or persistent; compare the ML-based ranking against a naive single-factor benchmark (e.g., momentum alone or composite fundamental score alone) to isolate whether the ML layer adds anything; and formally test the statistical significance of the alpha estimates (standard errors, t-stats) given the small sample size, rather than relying on the point estimate's sign alone.